Certified Robustness for Model-Free PID Tuning by Step-Response Inspection

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Abstract

The SPIN method tunes a PID controller without a model. After each setpoint step it counts the turns of the control error in three phase portraits and raises the gains until a count exceeds its limit. The limits were calibrated, and nothing guaranteed robustness. We prove two results. First, for any loop, the count of the integral portrait grows by exactly half a turn per setpoint crossing whenever the lobes of the error shrink. On loops that cancel the plant lags, which have a single gain parameter, the lobes always shrink in the relevant range, the sensitivity peak M_s grows strictly with the gain, and the error never crosses the setpoint if and only if $M_s \leq 1.394$. Second, for any stable digital loop, the increments of the error after a unit setpoint step are the impulse response of the sensitivity function. From one finite record we derive a lower and an upper bound on M_s that need no model and one assumption on the variation of the error after the record ends, and both converge to M_s. Used as the acceptance test of SPIN, the upper bound certifies every accepted tuning; the same record predicts which gain to lower, and a uniform backoff provably restores the target. Simulations on 19 loops confirm the theory.

1 Introduction

Let $L = CP$ be the loop transfer function and $S = 1/(1+L)$ the sensitivity function. The peak $M_s = \max_\omega |S|$ measures robustness: a value near 1.4 is safe, a value near 2 is close to the limit [2]. Tuning rules reach a chosen M_s with a model [2, 3]. After tuning, plants rarely check M_s, because a check needs a model or a test that disturbs production [5, 6].

1.1 The SPIN method

SPIN [1] tunes a PI or PID controller with gains K_i, K_p, K_d without a model. In each iteration it applies one setpoint step, records the control error e, and forms three phase portraits,

$$\Gamma_0 : (e, z), \quad \Gamma_1 : (\Delta e, e), \quad \Gamma_2 : (\Delta^2 e, \Delta e), \tag{1}$$

where z is the running sum of e and Δ the difference between samples. Portrait Γ_k reacts to the band of gain $k \in \{i, p, d\}$. The turn count N_k is the number of turns that Γ_k makes around its end point before the error stays within a band δ (2% of the step by default). Algorithm 1 gives the rule. SPIN needs no model and no plant-specific setting [1], but its limits $\bar{N}_k$ were chosen by calibration, and nothing guaranteed the robustness of the result. An online simulator lets the reader try SPIN.[1]

Algorithm 1: one iteration of SPIN [1]

```
Apply a setpoint step and record the error e.
if the record does not settle then
    divide K_i, K_p, K_d by 2, 4, 8                (back off)
else
    count the turns N_0, N_1, N_2 of Γ_0, Γ_1, Γ_2
    if N_k ≤ N̄_k for every band k then
        raise all gains by one step                (accept)
    else
        k ← lowest band with N_k > N̄_k
        lower the gain of band k by one step,
        raise the gains of the bands below k by one step
```

One step multiplies or divides a gain by $1/(1-\beta)$. Defaults: $\bar{N} = (0.5, 0.75, 1.0)$, $\delta = 0.02$, $\beta = 0.1$, each gain within $[0.001, 10]$ times its start.

1.2 Contributions

Result 1: what a count means (Section 2). Lemma 1 shows, for any loop, that a portrait turns by exactly half a turn between two zeros of its first coordinate, in any linear coordinates; so the count of Γ_0 is within half a turn of half the number of setpoint crossings when the lobes of the error shrink (Corollary 1). For loops that cancel the plant lags we prove that the lobes shrink, that M_s grows strictly with the loop gain, and that the error crosses the setpoint if and only if $M_s > 1.394$ (Propositions 1, 2). The Lambert-W and $1/e$ facts are classical [10, 11]; their link to the counts is new.

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Result 2: the step certifies M_s (Sections 3, 4). For a digital loop the record gives the sensitivity exactly (Lemma 2). Theorem 1 turns one finite record into certified lower and upper bounds on M_s. Theorem 2 makes them the acceptance test of SPIN: every accepted tuning satisfies $M_s \le M^*$, and the uniform back-off is stable and restores the target. Proposition 3 predicts from the same record which band to lower. Section 5 confirms all statements in simulation; noise and real loops are left to a follow-up.

1.3 Related work

Frequency responses have been computed from step and relay responses [7], sensitivity functions and margins from closed-loop data with reference excitation or designed experiments [8, 9]. The setpoint overshoot method tunes PI controllers from one setpoint step [14]. Model-free tuning by repeated experiments [12, 13] or from one data set [15] optimizes performance. Here one normal setpoint step, without special excitation, gives a certified bound on M_s inside a model-free rule.

2 Result 1: What a Turn Count Means

2.1 Half turns

In each portrait (1), the second coordinate q is the running sum of the first, p; in continuous time $\dot q = p$, and the turns are counted around the end point, which we place at the origin. We state the results in continuous time; the sampled counts of SPIN approximate them. For Γ_0 the zeros of $p = e$ are the setpoint crossings, for Γ_1 the extrema of e.

Lemma 1 (Half turns) *Let $\dot q = p$ with p continuous, and let $t_1 < t_2$ be consecutive zeros of p with $q(t_1)q(t_2) < 0$. On $[t_1, t_2]$ the curve (p, q) turns counterclockwise around the origin by exactly half a turn. So does its image under any linear map with positive determinant, such as the whitening of [1].*

Proof: On (t_1, t_2), p keeps one sign, so the curve stays in one open half-plane bounded by the q-axis, and q moves monotonically from $q(t_1)$ to $q(t_2)$, that is from one ray of the q-axis to the opposite ray. In the half-plane the angle lies in an open interval of length π, so it changes by exactly π; for $p > 0$, q increases, and for $p < 0$ it decreases, so the turn is counterclockwise in both cases. A linear map with positive determinant maps the half-plane to a half-plane and the two rays to opposite rays, and it keeps the orientation. □

Corollary 1 (Counts and crossings) *Let p change sign at $t_1 < \cdots < t_m$, $m \ge 1$, before the record settles, let $q(t_j)$ alternate in sign, and let $q(t_1)$ have the sign of p before t_1. Then the count of (p, q) satisfies $\frac{m-1}{2} < N < \frac{m+1}{2}$. For Γ_0, these conditions hold when the areas of the lobes of e between crossings decrease strictly.*

Proof: Each of the $m-1$ intervals adds π (Lemma 1). Before t_1 and after t_m the curve stays in one open half-plane and starts or ends on the ray from which it turns counterclockwise, so each end adds an angle in $(0, \pi)$. For Γ_0, $q(t_j) = -\int_{t_j}^{\infty} e\,dt$ is an alternating sum of lobe areas; when they decrease, it has the sign of its first term. □

2.2 The lag-cancelled loop

Lambda, IMC and SIMC tunings cancel the plant lags [3]: PI $K_c(1 + 1/(T_i s))$ with $T_i = T$ on $Ke^{-sL_d}/(Ts+1)$, or series PID with $T_i = T_1$, $T_d = T_2$ on a plant with two lags. The loop is then

$$L(s) = \frac{k\,e^{-sL_d}}{sL_d}, \qquad k = \frac{KK_cL_d}{T_i}. \tag{2}$$

Proposition 1 (Exact robustness) *The loop* (2) *is stable for $0 < k < \pi/2$, with gain margin $\pi/(2k)$ and poles $W_j(-k)/L_d$, where W_j are the branches of the Lambert W function [10]. Its peak M_s is a strictly increasing function of k, given by*

$$k = x\sin x - x^2\cos x, \qquad \frac{1}{M_s^2} = 1 - \frac{2k\sin x}{x} + \frac{k^2}{x^2}, \tag{3}$$

where $x = \omega_p L_d \in (0, \pi/2)$ is the normalized peak frequency.

Proof: The phase of L is $-\pi/2 - x$; it reaches $-\pi$ at $x = \pi/2$, where $|L| = 2k/\pi$. The poles solve $sL_de^{sL_d} = -k$. Let $h(x) = |1+L|^2 = 1 - 2k\sin x/x + k^2/x^2$. Then $h'(x) = 2k(\varphi(x) - k)/x^3$ with $\varphi(x) = x\sin x - x^2\cos x$, which increases on $(0, \pi)$ with $\varphi(\pi/2) = \pi/2 > k$. So h has a single minimum $x^* \in (0, \pi/2)$ on $(0, \pi)$. On $[\pi, 2\pi]$, $h \ge 1$; for $x \ge 2\pi$, $h \ge (1 - k/x)^2 > (1 - 2k/\pi)^2 = h(\pi/2)$. So x^* is the global minimum, which gives (3). By the envelope theorem, $\frac{d}{dk}h(x^*) = \frac{2}{x^{*2}}(k - x^*\sin x^*) = -2\cos x^* < 0$. □

SIMC with $\tau_c = \theta = L_d$ and $T \le 8\theta$ cancels the lag and gives $k = \frac{1}{2}$, so GM $= \pi$ and $M_s = 1.59$ by (3), exactly the values in [4, Table 2].

After a unit setpoint step, $e = 1$ on $[0, L_d]$ and $\dot e(t) = -(k/L_d)\,e(t - L_d)$. Let $c_1 < c_2 < \ldots$ be the setpoint crossings, $c_0 = 0$, lobe i the interval $[c_i, c_{i+1}]$, and $E_0 = 1$.

Proposition 2 (Crossings) *For the loop* (2) *with $0 < k < \pi/2$: (a) the error never crosses the setpoint if and only if $k \le 1/e$, that is $M_s \le M_s(1/e) = 1.394$; for $k > 1/e$ it crosses infinitely often. (b) Lobe i contains exactly one extremum E_i, at $c_i + L_d$. (c) $|E_{i+1}| \le r(k)\,|E_i|$, with $r(k) = k^2/2$ for $k \le 1$ and $r(k) = k - \frac{1}{2}$ for $k \ge 1$. (d) The lobe areas are $\frac{L_d}{k}(|E_i| + |E_{i+1}|)$. For $k < 3/2$ they decrease strictly, so Corollary 1 holds for Γ_0: N_0 is within half a turn of half the number of crossings.*

Proof: (a) The record is the fundamental solution of the delay equation. It is positive if and only if $s + (k/L_d)e^{-sL_d} = 0$ has a real root, that is $k \le 1/e$, and for $k > 1/e$ all solutions oscillate [11]; M_s follows from Proposition 1. (b) $\dot e(t)$ has the sign of

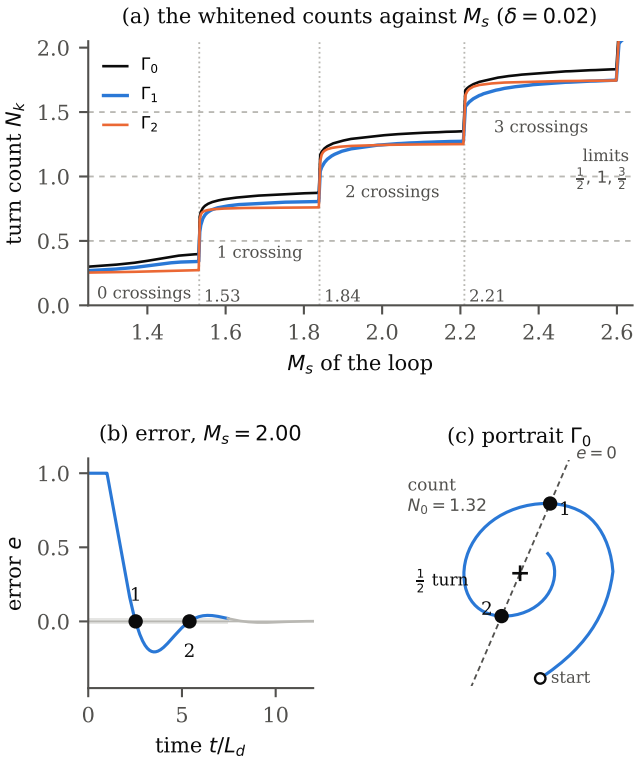


Figure 1: Lemma 1 and Proposition 2, evaluated on the lag-cancelled loop. (a) The three whitened counts against M_s ($\delta = 0.02$); N_0 steps by half a turn at each setpoint crossing before settling (Corollary 1). (b) A record with $M_s = 2.00$ crosses twice before it settles within δ. (c) Its portrait Γ_0: between crossings (on the dashed line $e = 0$) the curve turns exactly half a turn around the end point ($+$). The radius is compressed for visibility; the angle is not.

$-e(t-L_d)$. By induction over the lobes, e moves away from zero on (c_i, c_i+L_d), where $e(t-L_d)$ lies in lobe $i-1$, so lobe i is longer than L_d; and e moves toward zero on $(c_i+L_d, c_{i+1}+L_d)$, where $e(t-L_d)$ lies in lobe i. For $i=0$ the extremum is the plateau $e=1$ on $[0,L_d]$. (c) On lobe i, $|e| \le |E_i|$. After the extremum, $|\dot{e}| \le (k/L_d)|E_i|$, so e needs at least L_d/k to reach zero and $|e(t)| \le |E_i| \min\{1, k(c_{i+1}-t)/L_d\}$ on the last L_d of the lobe. Since $e(c_{i+1}) = 0$, $E_{i+1} = -\frac{k}{L_d}\int_{c_{i+1}-L_d}^{c_{i+1}} e\,dt$, and the bound integrates to $r(k)|E_i|$. (d) Integrate $\dot{e}$ from c_i+L_d to $c_{i+1}+L_d$; the extrema alternate in sign. By (c), $|E_{i+2}| < |E_i|$ for $r(k) < 1$, that is $k < 3/2$, where $M_s = 26.2$. □

So on lag-cancelled loops N_0 reads the number of crossings, and crossings read M_s: none if and only if $M_s \le 1.394$, and by (c) the extrema decay at least geometrically. Fig. 1 illustrates both statements.

2.3 Counts do not certify in general

Outside lag-cancelled loops, crossings and M_s are not tied. A first-order plant with $T = 0.3L_d$ under PI with $T_i = 8T$ and gain margin 1.624 has $M_s = 2.62$, yet its error never crosses the setpoint by more than 2%; the oscillation rides on a slow decay. Conversely, $e^{-s}/(20s+1)$ with PI ($K_c = 2.5$, $T_i = 5$) crosses twice with 26% overshoot, yet $M_s = 1.34$; the overshoot comes from the controller zero. These values are computed for the continuous loops. So counts can guide tuning but cannot certify it. The same step, however, contains M_s.

3 Result 2: The Same Step Certifies M_s

A digital controller runs at sampling period T_s. The loop it closes, seen at the samples, is linear and time invariant with loop transfer function $L_d(z)$ and sensitivity $S_d = 1/(1+L_d)$, which is the sensitivity of the loop that actually runs; we write $M_s = \max_{\theta\in[0,\pi]} |S_d(e^{j\theta})|$, $\theta = \omega T_s$.

Lemma 2 (Exact identity) *Let the loop be stable, with one degree of freedom and strictly proper L_d (the measurement precedes the control action), and let e_n be the error after a unit setpoint step at $n = 0$ from rest, without disturbances, so $e_{-1} = 0$ and $e_0 = 1$; let $a_n = e_n - e_{n-1}$. Then*

$$S_d(e^{j\theta}) = \sum_{n\ge 0} a_n e^{-jn\theta}. \tag{4}$$

Proof: $E(z) = S_d(z)R(z)$ with $R(z) = 1/(1-z^{-1})$, so $S_d = (1-z^{-1})E$. In a stable loop a_n decays exponentially, so the series converges on the unit circle. □

In continuous time the same argument gives $S(j\omega) = \int_0^\infty e^{-j\omega t}de(t)$. With two degrees of freedom, $u = C_r r - C_{fb}y$, it applies to u and y and gives the plant, $P_d = \sum \Delta y_n e^{-jn\theta} / \sum \Delta u_n e^{-jn\theta}$ where the denominator is nonzero, and $S_d = 1/(1+C_{fb}P_d)$; we claim no bound for this form. A record is finite. The only quantity that it cannot give is the variation of the error after it ends.

Assumption 1 (Settled record) *The loop is stable, and the record $e_0, \dots, e_N$ satisfies $\sum_{n>N} |a_n| \le \tau$ for a known τ.*

In a stable loop $|a_n| \le c\rho^n$ with $\rho < 1$, so the tail variation falls geometrically with the record length; τ states how settled the record is, in units of the step.

Theorem 1 (Certified bounds from one record) *Under Assumption 1, let $\hat{S}(\theta) = \sum_{n=0}^{N} a_n e^{-jn\theta}$, let $\theta_k = kh$, $k = 0, \dots, K$, be a grid with $Kh = \theta_1 \le \pi$, and let*

$$J = \sum_{n=0}^{N} |n - n_c|\,|a_n|, \quad B = \sum_{n=1}^{N-1} |a_{n+1} - a_n| + |a_N| + 2\tau,$$

for any n_c. Then $\underline{M} \le M_s \le \overline{M}$, with

$$\underline{M} = \max_k |\hat{S}(\theta_k)| - \tau,$$
$$\overline{M} = \max\left\{ \max_k |\hat{S}(\theta_k)| + \tfrac{hJ}{2} + \tau,\ 1 + \tfrac{B}{\sin(\theta_1/2)} \right\}, \tag{5}$$

where the second term is dropped if $\theta_1 = \pi$. With τ equal to the true tail, $N \to \infty$, $h \to 0$ and $\theta_1 = \pi$, both bounds converge to M_s.

Algorithm 2: certified SPIN: line 7 of Algorithm 1 becomes

7a: compute $\overline{M}$ by Theorem 1: θ_1 solves $1+B/\sin(\theta_1/2)=M^*$ ($\theta_1=\pi$ if none), $hJ/2\le\eta$
7b: **if** $\overline{M}\le M^*$ **then** raise all gains by one step (accept)
7c: **else** predict, by Proposition 3(d), M_s after SPIN's move of each band (lower it, raise those below)
7d: apply the highest band predicted to lower M_s, at most $n_{\max}$ times in a row; otherwise lower all gains by one step; end the iteration

Defaults: $M^*=1.7$, $\tau=10^{-3}$, $\eta=2\cdot10^{-3}$, $n_{\max}=10$.

Proof: By (4), $|S_d-\hat{S}|\le\sum_{n>N}|a_n|\le\tau$ at every θ, which gives $\underline{M}$. On $[0,\theta_1]$, $|\hat{S}|$ equals $|e^{jn_c\theta}\hat{S}|$, whose derivative is bounded by J; every θ is within $h/2$ of a grid point, which gives the first term of $\overline{M}$. On $[\theta_1,\pi]$, let $z=e^{-j\theta}$ and $Z_n=\sum_{m=1}^{n}z^m$, so $|Z_n|\le 2/|1-z|=1/\sin(\theta/2)$. Summation by parts gives $\sum_{n\ge1}a_nz^n=\sum_{n\ge1}(a_n-a_{n+1})Z_n$, and $\sum_{n\ge N}|a_n-a_{n+1}|\le|a_N|+2\tau$, so $|S_d-a_0|\le B/\sin(\theta/2)$ with $a_0=1$. Convergence: the true tail and hJ tend to zero. □

Every quantity in (5) except τ comes from the record; the grid values are one chirp z-transform. The second term needs no grid: for a smooth record B is small and certifies all high frequencies at once. Measurement noise is outside this paper.

4 Certified SPIN

Algorithm 2 changes only line 7 of SPIN. The counts still shape the tuning; the bound decides whether it is accepted.

Theorem 2 (Certified SPIN) *Let the plant be stable, let $K_iP_d(1)>0$, and let Assumption 1 hold for every record that reaches line 7a. (i) Every tuning accepted by Algorithm 2 has $M_s\le M^*$. (ii) For gains K, let $m=\min\{0,\inf_{0<\theta\le\pi}\mathrm{Re}\,L_d(e^{j\theta})\}$. For $0<\lambda<1/|m|$ the loop with gains λK is stable and*

$$M_s(\lambda K)\le\frac{1}{1-\lambda|m|}. \tag{6}$$

(iii) If $M^*>1+2\tau+\eta$, then after $j\ge\ln\frac{|m|M'}{M'-1}/\ln\frac{1}{1-\beta}$ uniform back-offs from K, $M'=M^*-2\tau-\eta$, the test of line 7b passes.

Proof: (i) Theorem 1. (ii) L_d is bounded on the unit circle except near $z=1$, where the integral term gives a bounded real part because P_d is analytic there and $\mathrm{Re}\,1/(e^{j\theta}-1)=-\frac{1}{2}$; so m is finite. Then $|1+\lambda L_d|\ge1+\lambda\,\mathrm{Re}\,L_d\ge1-\lambda|m|>0$, which gives (6). The Nyquist curve of λL_d lies in $\mathrm{Re}\ge-\lambda|m|>-1$, also on the indentation at $z=1$, where $K_iP_d(1)>0$ makes the real part large and positive; it cannot encircle -1, and L_d has no poles outside the unit disc, so the loop is stable. (iii) After j back-offs, $\lambda=(1-\beta)^j$, and (6) gives $M_s\le M'$. In (5) the grid term is then at most $M_s+\tau+\eta+\tau\le M^*$, and the second term is at most M^* by the choice of θ_1. □

Theorem 2 needs no gain box: safety comes from the bound, not from limits on the gains. Part (iii) is a statement about the bound; whether line 7b is reached also depends on the counts of line 6. Lowering a band can be cheaper than lowering all gains, and the same record tells which band helps. Let $c_k=K_k\,\partial C/\partial K_k$ for $k=i,p,d$, with the derivative filter tied to K_d/K_p, so that $\sum_k c_k=C$, and let $T=1-S_d$.

Proposition 3 (Shaping from one step) *Let $|S_d|$ have a unique peak at θ_p. Then: (a) $g_k=\partial\ln M_s/\partial\ln K_k=-\mathrm{Re}(Tc_k/C)$ at θ_p, and $\sum_k g_k=\mathrm{Re}\,S_d(e^{j\theta_p})-1$. (b) Lowering all gains together lowers M_s to first order if and only if $\mathrm{Re}\,S_d>1$ at θ_p, that is $|L_d+\frac{1}{2}|<\frac{1}{2}$. (c) SPIN's move of band k lowers M_s to first order if and only if $g_k>\sum_{j<k}g_j$; if $\sum_k g_k>0$, such a band exists. (d) For any controller C' on the same plant, $S_d'=S_d/(S_d+(1-S_d)C'/C)$ exactly. If the record gives $\hat{S}=S_d+\varepsilon$, the prediction $\hat{S}'$ errs by $\hat{S}'-S_d'=\varepsilon\,(C'/C)\,S_d'\hat{S}'/(S_d\hat{S})$.*

Proof: (a) $\partial\ln S_d/\partial\ln K_k=-Tc_k/C$; at a unique peak $\ln M_s=\mathrm{Re}\ln S_d$ (Danskin), and $\sum c_k=C$. (b) is $\sum g_k>0$, and $\mathrm{Re}\frac{1}{1+L}>1\Leftrightarrow|L+\frac{1}{2}|<\frac{1}{2}$. (c) The move changes $\ln M_s$ by $\ln\frac{1}{1-\beta}(\sum_{j<k}g_j-g_k)$. With $P_k=\sum_{j\le k}g_j$ the condition is $P_k>2P_{k-1}$; it holds for the smallest k with $P_k>0$. (d) $L_d'=L_dC'/C$ with $L_d=(1-S_d)/S_d$; subtracting the two expressions gives the error. □

By (b), where $\mathrm{Re}\,S_d<1$ at the peak a small uniform back-off raises M_s, although a large one restores it (Theorem 2(ii)). Safety never depends on the prediction: the next step checks each move.

5 Confirmation in Simulation

Setup We compare SPIN as published with certified SPIN (Algorithm 2), with 200 iterations, derivative filter ratio 10 and the gain box $[0.001,10]$ of [1]. Because the box ties the result to the start, every loop runs from two starts: SIMC on a half-rule model with $\tau_c=3\theta$, and twice as bold ($\tau_c=\theta$). The plants are those of [1],

$$P_1=\frac{e^{-s}}{(10s+1)(s+1)^3},\qquad P_2=\frac{1.25\,e^{-8s}}{(5s+1)^4},$$
$$P_3=\frac{e^{-10s}}{(2s+1)(s+1)^3},\qquad P_4=\frac{e^{-4s}}{(8s+1)^6},$$

with PI and PID; a delay-dominant $e^{-5s}/(s+1)$, two lag-dominant $e^{-s}/(10s+1)$ and $e^{-0.5s}/((10s+1)(s+1))$ with PI; a fast $e^{-0.1s}/((s+1)(0.5s+1))$ with PID; and $e^{-L_ds}/((s+1)(0.5s+1)(0.25s+1))$ with PID and $L_d\in\{0.25,\dots,16\}$ (dead-time share 0.125 to 0.90). The controller runs at $T_s=\min(T_a/100,T_f/20)$ for PID, with T_f the derivative filter time constant and T_a the plant's time scale, and at $T_a/1000$ for PI; each lag is discretized exactly for a piecewise-constant input and the delay is rounded to whole samples. The record covers $15\,T_a$. All M_s below are those of this digital loop, computed

from its exact L_d. At the end, SPIN cycles between a few tunings; we report the most aggressive one accepted in the last 60 iterations.

Result 1 On the lag-cancelled loop with $1/e < k < 3/2$, the bound (c) of Proposition 2 held for every extremum (with equality for the first when $k \geq 1$), the lobe areas decreased, each interval between crossings turned by $\frac{1}{2}$ turn to 10^{-6}, also after a linear map, and the whitened count of SPIN satisfied Corollary 1 for all 126 values of k tested; (3) matched a direct search of M_s to six digits.

Lemma 2 and Theorem 1 For each of the 146 tunings accepted in the six runs (distinct within each run), we simulated a record four times longer to measure the true tail $\sum_{n>N} |a_n|$. It was at most $2.8 \cdot 10^{-5}$, so Assumption 1 held with $\tau = 10^{-3}$; the identity (4) held up to this tail and round-off; and every true M_s lay within its bounds (here θ_1 is set at the level of the read peak), with $\overline{M} - M_s \leq \eta + \tau + 2.8 \cdot 10^{-5}$. On short records the tail is large and the bounds are wide, yet they contain M_s and close in as the error settles (Fig. 2(b)).

Theorem 2 and Proposition 3 Published SPIN ended above $M^* = 1.7$ on 9 of the 19 loops from the cautious start (up to 2.01) and on 8 from the bold start (up to 2.09); which loops failed depended on the start, because the gain box often set the result (Table 1). Certified SPIN ended at or below M^* on all loops from both starts, as Theorem 2(i) guarantees; the highest value was 1.69. Without the gain box, published SPIN went above the target on 12 of the 18 loops that converged (up to 2.05), and certified SPIN stayed at or below it on all 18 (up to 1.68); the family at $L_d = 16$ converged under neither rule without the box. Along the uniform back-off from the most aggressive published tuning of each loop (cautious start), the loop stayed stable and (6) held; Proposition 3(a) held to 10^{-7}, and the prediction (d) of the M_s after every band move matched the true value to 10^{-7} (Fig. 3(b)). The price of the target is load rejection: the integrated absolute error after a unit load step at the plant input rose by 7.8% on average over the 19 loops (6.4% from the bold start).

6 Use in Practice

By eye, on a lag-cancelled loop: an error that never crosses the setpoint means $M_s \leq 1.394$; SIMC crosses (Proposition 2). *By computer:* every setpoint change in a historian, made from rest and without disturbances, certifies M_s by Theorem 1, without a model. *Scope:* the theorems assume a linear loop and a settled record (Assumption 1); noise in the record and in the counts and real loops are the subject of a follow-up. Scripts that reproduce every number and figure are in `https://github.com/ottapav/robopid-simulator`, folder `docs/ECC27_certified_SPIN`.

Table 1: M_s of the digital loop for the most aggressive tuning that published and certified SPIN accept, from the cautious (3θ) and the bold (θ) start, the band that owns the peak of the published tuning, and the change of load-step IAE (cautious start).

		published		certified			
		3θ	θ	3θ	θ	owner	ΔIAE
P_1	PI	1.46	1.46	1.46	1.46	Γ_1	–
	PID	1.66	**1.84**	1.66	1.61	Γ_1	–
P_2	PI	1.64	1.67	1.64	1.67	Γ_1	–
	PID	1.59	**1.78**	1.59	1.66	Γ_1	–
P_3	PI	1.67	1.63	1.67	1.63	Γ_0	–
	PID	**1.81**	**1.74**	1.69	1.60	Γ_1	+23%
P_4	PI	1.55	1.60	1.55	1.60	Γ_1	–
	PID	1.61	1.55	1.61	1.55	Γ_1	–
delay-dom.	PI	1.68	1.63	1.68	1.63	Γ_0	–
lag-dom.	PI	**2.01**	1.69	1.67	1.69	Γ_1	+52%
2 lags	PI	1.44	1.47	1.44	1.47	Γ_1	–
fast	PID	**1.89**	**1.98**	1.66	1.69	Γ_2	0%

7 Conclusion

The count of Γ_0 is within half a turn of half the number of setpoint crossings when the lobes of the error shrink; on lag-cancelled loops the crossings read M_s exactly. On any stable digital loop the setpoint step that SPIN already records gives the sensitivity function exactly, and one finite record gives certified bounds on M_s. As the acceptance test of SPIN, the upper bound certifies every accepted tuning, and a uniform back-off provably restores the target. Simulation confirmed each statement.

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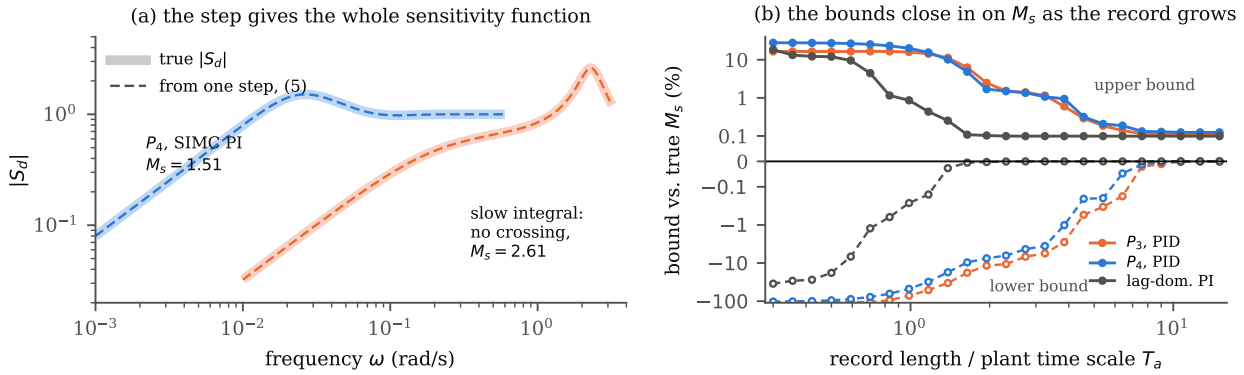


Figure 2: Lemma 2 and Theorem 1. (a) $|S_d|$ from one setpoint step (dashed) lies on the true one (thick), for P_4 with SIMC PI and for the slow-integral loop whose peak the counts miss. (b) The bounds (5) relative to the true M_s against the record length, with τ the true tail, for three published tunings; the floor of the upper bound is η.

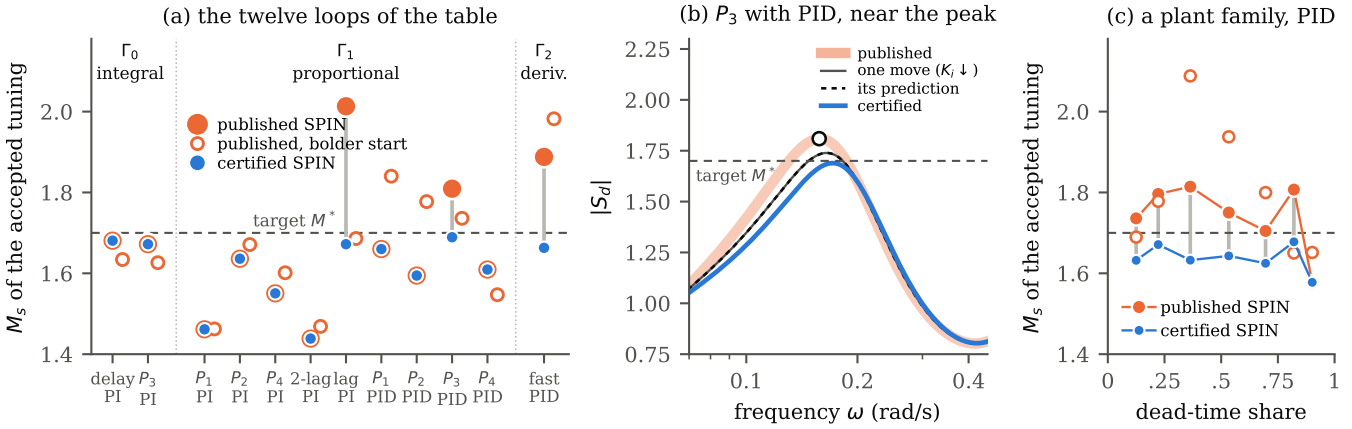


Figure 3: Published against certified SPIN ($M^* = 1.7$). (a) The loops of Table 1, grouped by the band that owns the peak of the published tuning. Filled orange: published, cautious start; open orange: published, bold start; blue: certified, cautious start. (b) P_3 with PID near the peak. Published tuning ($\overline{M} = 1.81 > M^*$, ○); one move lowering K_i ($M_s = 1.739$, as predicted from the same record by Proposition 3(d)); the tuning certified SPIN accepts (1.689). (c) The family $e^{-L_d s}/((s+1)(0.5s+1)(0.25s+1))$ with PID against the dead-time share.

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